

# Complex Analysis

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based on course by

We will study holomorphic and analytic functions  $f : \mathbb{C} \rightarrow \mathbb{C}$ . We will follow Sterin-Shakarchi's Princeton Lecture Series.

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## 1. Trailer of shit we will see?!

### Definition: Holomorphic (differentiation definition)

A function is said to be holomorphic at a point  $z \in \mathbb{C}$  if

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists in all the directions and agrees. In that case,  $\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \rightarrow f'(z)$ .

### Definition: Holomorphic (power series definition)

A function is holomorphic if  $f(z) = \sum_{n=0}^{\infty} a_n z^n$

We can obviously view  $f : \mathbb{C} \rightarrow \mathbb{C}$  as  $f_1 : \mathbb{R} \rightarrow \mathbb{R}$  and  $f_2 : \mathbb{R} \rightarrow \mathbb{R}$ . Also,

**Theorem 1.1.** *If a function is differentiable once at a point, it is differentiable infinitely.*

Given a path  $\gamma$  in complex plane, we can integrate over it from  $z_1, z_2 \in \gamma$  by finding a differentiable and continuous function  $h : [a, b] \rightarrow \mathbb{C}$  such that  $h(a) = z_1$  and  $h(b) = z_2$  then

$$\int_{\gamma} f(x) dz = \int_a^b f(h(t)) h'(t) dt$$

**Cauchy's Theorem 1.2.** *Given a closed curve  $\gamma$  in  $\mathbb{C}$ ,*

$$\int_{\gamma} f(x) dz = 0$$

**Cauchy's Integral Formula 1.3.** Given  $f$ 's value on the boundary of a disc  $C$  (the disk is named  $D$  and boundary is  $C$ ) then:

$$f(z) = \int_C \frac{f(\zeta)}{\zeta - z}, z \in D$$

**Corollary 1.4.**

$$f^{(n)}(z) = \frac{n!}{2\pi} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}$$

### Definition: Entire Function

An Entire function that is holomorphic in  $\mathbb{C}$

**Liouville 1.5.** A bounded entire function is constant.

**Schwarz Reflection Principle 1.6.**

**TODO.** Google Karlo bhai

## 2. Preliminaries

$z \in \mathbb{C}$